

ETHAN PARK

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EDUCATION

University of Waterloo

Bachelor of Applied Science, Nanotechnology Engineering

- Recipient of University of Waterloo President's Scholarship

Sep 2023 - Apr 2028

Waterloo, ON

EXPERIENCE

Cell Engineering Intern

National Research Council Canada

Sep 2025 - Apr 2026

Mississauga, ON

- Delivered 40x increase in Li-ion cathode slurry throughput by programming a liquid-dispensing robot and 3D printing custom labware for seamless integration into a self-driving lab (SDL).
- Assembled and characterized 2000+ Li-ion half cells using CV, EIS, and OCP, and accelerated data analysis by 36x using Python to automate data visualization, noise reduction, and peak finding.
- Investigated NMC phase compositions (XRD), oxygen-assisted PCAM calcination, and Niobium/Tungsten doping concentrations using Bayesian optimization to achieve specific capacities within 90% of literature values.

Research Assistant, Redox Flow Battery Design

Van Der Heijden Lab

Apr 2025 - Aug 2025

Waterloo, ON

- Improved mass transport across electrode interfaces by 50% using SolidWorks CFD to optimize flow field and inlet designs.
- Reduced electrolyte leakage and mechanical warping by 90% in 3D printed flow cells through iterative optimization of SLA printing parameters, resin selection, and curing processes.
- Investigated nitrogen-assisted carbonization conditions to convert highly porous polymer structures to functional electrodes.

Electrical Engineering Intern

Smith + Andersen

Jan 2025 - Apr 2025

Toronto, ON

- Drafted electrical device and power distribution drawings in AutoCAD for 15+ public infrastructure projects.
- Completed electrical load and lighting calculations to support energy-efficient layouts, reducing client expenses by 20%.
- Conducted engineering site audits to monitor project progress and incorporated P.Eng. review feedback to validate CAD deliverables for contractors.

EXTRACURRICULAR

Semiconductor Process Development Team

Waterloo Hacker Fab

May 2026 - Present

Waterloo, ON

- Designed a test bench for I-V characterization of fabricated semiconductor devices to support electrical testing.
- Achieved 3 micron patterning resolution through spin coating and lithography process development.
- Investigating thermal oxidation parameters using a custom tube furnace to optimize silicon oxide growth for microfabrication processes and improve process repeatability.

Class Representative

Waterloo Engineering Society

Sep 2023 - Jan 2025

Waterloo, ON

- Elected to represent the Nanotechnology cohort, serving as a key liaison between students and society executives
- Gained strong communication and presentation skills by contributing budgetary solutions during society discussions
- Engaged hundreds of engineering students across campus by collaborating with fellow society reps to organize and promote large-scale student social events

TECHNICAL SKILLS

Characterization Techniques: XRD, SEM, EIS, CV, OCP, UV-Vis, IV

Battery Processing: NMC Cathode Synthesis, Slurry Compounding, Electrode Fabrication, Cell Assembly

Micro Fabrication: Photolithography, Spin Coating, Thin Film Growth, Thermal Oxidation, Cleanroom ISO 4

Programming: Python (NumPy, SciPy, Pandas), Matlab, Git

CAD and Simulation: SolidWorks, COMSOL, AutoCAD, Simulink